

AGH University of Krakow

Mobile App Development

Laboratory 5 Report

Static Layouts in Android

Student:	Carlos Guzman
Project name:	CarlosGuzmanLayoutsAndActions
Language:	Kotlin
Development environment:	Android Studio
Date:	April 22, 2026

1. Objective

The objective of this laboratory practice is to create static user interfaces in Android Studio using XML layout files. During the practice, different types of layouts were explored, including `RelativeLayout` and `LinearLayout`, as well as the use of interface resources such as colors, strings, and drawable images.

2. General Description

In this laboratory, an Android application project was created in Kotlin. The practice focused on editing the XML files directly in order to understand how static layouts are defined in Android. Different interface elements such as text fields, images, and buttons were positioned inside the layouts. In addition, custom colors and string resources were declared in the corresponding files under the `res` directory.

At the current stage, the project includes the following layout files:

- `activity_main.xml`
- `layout_weekend.xml`
- `layout_calculator.xml`

3. Resource Files

3.1. `colors.xml`

The following file defines the colors used in the layouts.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <resources>
3     <color name="black">#FF000000</color>
4     <color name="white">#FFFFFFFF</color>
5     <color name="layout_background">#2196F3</color>
6 </resources>
```

3.2. `strings.xml`

The following file defines the text resources used by the application.

```

1 <resources>
2     <string name="app_name">CarlosGuzmanLayoutsAndActions</string>
3     <string name="hello_string">Hello There&#10;From XML File</string>
4     <string name="android_is_so_awesome">Android is so&#10;awesome</string>
5     <string name="remeber">Remember!</string>
6 </resources>

```

4. Main Layout: activity_main.xml

The first part of the practice consists of creating a static screen using a `RelativeLayout`. A custom background color was added and two text elements were positioned on the screen. These elements were configured using XML attributes such as width, height, text size, text style, alignment, margins, and background colors.

This layout was used to understand how elements can be arranged relative to the parent container and to each other.

4.1. XML Code

```

1 <?xml version="1.0" encoding="utf-8"?>
2 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:id="@+id/main"
4     android:layout_width="match_parent"
5     android:layout_height="match_parent"
6     android:background="@color/layout_background">
7
8     <EditText
9         android:id="@+id/hello_there_edittext"
10        android:layout_width="140dp"
11        android:layout_height="wrap_content"
12        android:text="@string/hello_string"
13        android:textSize="18sp"
14        android:textStyle="bold"
15        android:background="@android:color/holo_green_light"
16        android:textColor="@android:color/black"
17        android:gravity="center"
18        android:padding="6dp"
19        android:layout_centerHorizontal="true"
20        android:layout_centerVertical="true"

```

```

21     android:layout_marginEnd="4dp" />
22
23 <EditText
24     android:id="@+id/awesome_android_edittext"
25     android:layout_width="140dp"
26     android:layout_height="wrap_content"
27     android:text="@string/android_is_so_awesome"
28     android:textSize="18sp"
29     android:textStyle="bold"
30     android:background="@android:color/holo_orange_dark"
31     android:textColor="@android:color/black"
32     android:gravity="center"
33     android:padding="6dp"
34     android:layout_toEndOf="@id/hello_there_edittext"
35     android:layout_alignTop="@id/hello_there_edittext" />
36
37 </RelativeLayout>

```

4.2. Screenshot

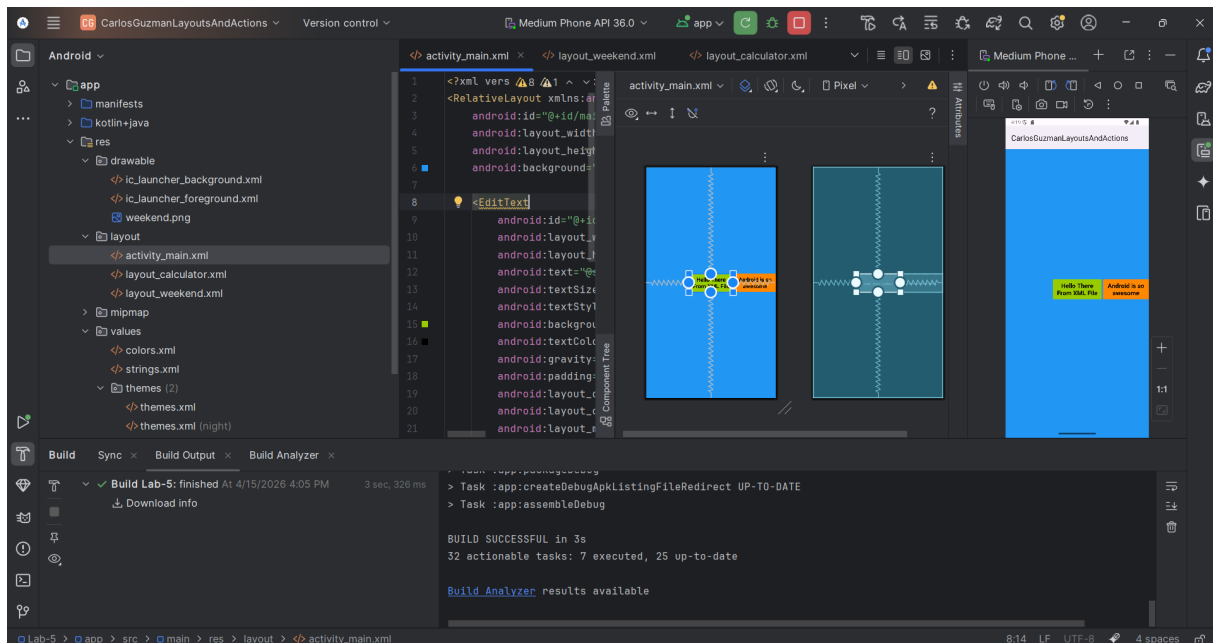


Figure 1: Main layout shown in the emulator.

5. Weekend Layout: layout_weekend.xml

The second part of the practice consists of creating a new XML layout file using a **LinearLayout**. In this screen, a title text and an image resource were added. This part of the laboratory helped to understand how to import drawable resources and how to organize interface elements vertically.

The image was added to the **drawable** folder and referenced inside the XML file. This layout is intended to show a simple reminder-style screen with a title and an image underneath.

5.1. XML Code

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:id="@+id/main"
4     android:orientation="vertical"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent">
7
8     <EditText
9         android:layout_width="match_parent"
10        android:layout_height="wrap_content"
11        android:text="@string/remeber"
12        android:textSize="24sp"
13        android:textStyle="bold"
14        android:gravity="center"
15        android:background="@android:color/holo_orange_dark"
16        android:textColor="@color/white"
17        android:padding="12dp" />
18
19    <ImageView
20        android:layout_width="match_parent"
21        android:layout_height="0dp"
22        android:layout_weight="1"
23        android:src="@drawable/weekend"
24        android:scaleType="centerCrop"
25        android:contentDescription="Weekend image" />
26
27 </LinearLayout>
```

5.2. Screenshot

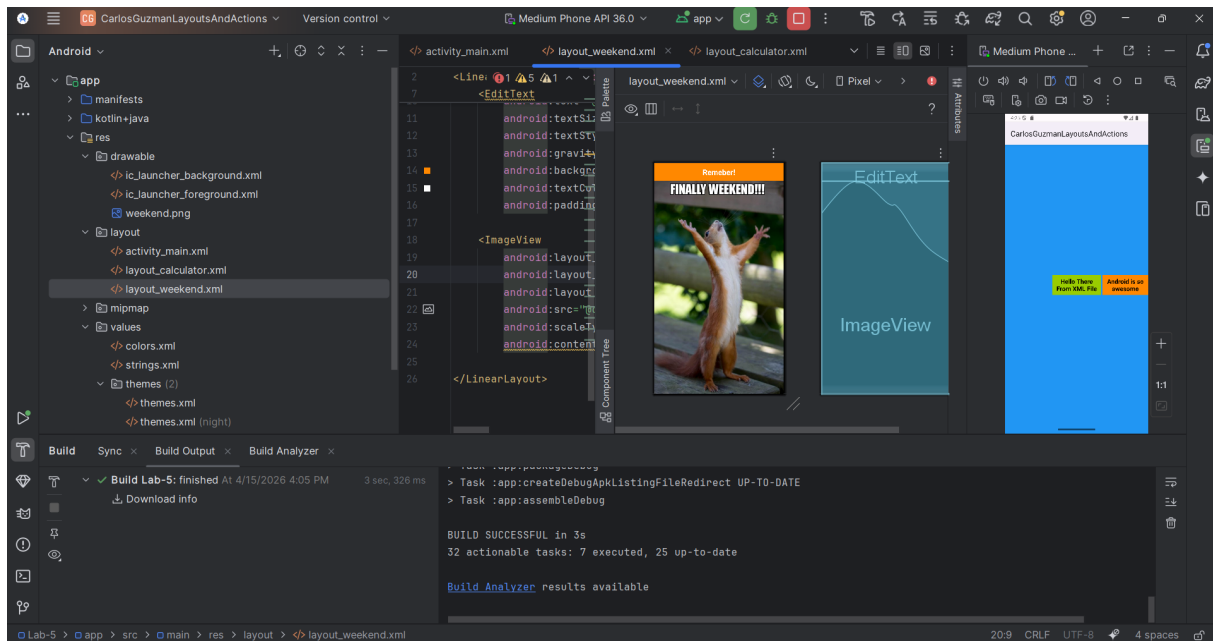


Figure 2: Weekend layout shown in the emulator.

6. Calculator Layout: layout_calculator.xml

The last part developed so far is a calculator-style interface made with XML. This layout was designed using nested `LinearLayout` containers in order to simulate rows of buttons and a display area at the top. It includes number buttons and operator buttons arranged in a grid-like structure.

This section demonstrates the use of multiple nested layouts, spacing, alignment, button styling, and a display area for the calculator.

6.1. XML Code

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
3     android:id="@+id/main"
4     android:orientation="vertical"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent"
7     android:background="#FFFFFF"
8     android:padding="16dp">
9
10    <EditText
```

```

11         android:id="@+id/display"
12         android:layout_width="match_parent"
13         android:layout_height="wrap_content"
14         android:text="5555"
15         android:textSize="48sp"
16         android:textStyle="bold"
17         android:gravity="end"
18         android:padding="16dp"
19         android:inputType="none"
20         android:background="@android:color/white" />
21
22     <View
23         android:layout_width="match_parent"
24         android:layout_height="1dp"
25         android:background="#CCCCCC"
26         android:layout_marginBottom="16dp" />
27
28     <LinearLayout
29         android:layout_width="match_parent"
30         android:layout_height="wrap_content"
31         android:orientation="horizontal"
32         android:gravity="center"
33         android:layout_marginBottom="8dp">
34
35         <Button
36             android:layout_width="70dp"
37             android:layout_height="70dp"
38             android:text="1"
39             android:backgroundTint="#7B1FA2"
40             android:textColor="@color/white"
41             android:layout_margin="4dp"
42             android:textSize="18sp" />
43
44         <Button
45             android:layout_width="70dp"
46             android:layout_height="70dp"
47             android:text="2"
48             android:backgroundTint="#7B1FA2"
49             android:textColor="@color/white"
50             android:layout_margin="4dp"
51             android:textSize="18sp" />

```

```

52
53     <Button
54         android:layout_width="70dp"
55         android:layout_height="70dp"
56         android:text="3"
57         android:backgroundTint="#7B1FA2"
58         android:textColor="@color/white"
59         android:layout_margin="4dp"
60         android:textSize="18sp" />
61
62     <Button
63         android:layout_width="70dp"
64         android:layout_height="70dp"
65         android:text="/"
66         android:backgroundTint="#7B1FA2"
67         android:textColor="@color/white"
68         android:layout_margin="4dp"
69         android:textSize="18sp" />
70 </LinearLayout>
71
72 <LinearLayout
73     android:layout_width="match_parent"
74     android:layout_height="wrap_content"
75     android:orientation="horizontal"
76     android:gravity="center"
77     android:layout_marginBottom="8dp">
78
79     <Button
80         android:layout_width="70dp"
81         android:layout_height="70dp"
82         android:text="4"
83         android:backgroundTint="#7B1FA2"
84         android:textColor="@color/white"
85         android:layout_margin="4dp"
86         android:textSize="18sp" />
87
88     <Button
89         android:layout_width="70dp"
90         android:layout_height="70dp"
91         android:text="5"
92         android:backgroundTint="#7B1FA2"

```



```

93         android:textColor="@color/white"
94         android:layout_margin="4dp"
95         android:textSize="18sp" />
96
97     <Button
98         android:layout_width="70dp"
99         android:layout_height="70dp"
100         android:text="6"
101         android:backgroundTint="#7B1FA2"
102         android:textColor="@color/white"
103         android:layout_margin="4dp"
104         android:textSize="18sp" />
105
106     <Button
107         android:layout_width="70dp"
108         android:layout_height="70dp"
109         android:text="-"
110         android:backgroundTint="#7B1FA2"
111         android:textColor="@color/white"
112         android:layout_margin="4dp"
113         android:textSize="18sp" />
114 </LinearLayout>
115
116 <LinearLayout
117     android:layout_width="match_parent"
118     android:layout_height="wrap_content"
119     android:orientation="horizontal"
120     android:gravity="center"
121     android:layout_marginBottom="8dp">
122
123     <Button
124         android:layout_width="70dp"
125         android:layout_height="70dp"
126         android:text="7"
127         android:backgroundTint="#7B1FA2"
128         android:textColor="@color/white"
129         android:layout_margin="4dp"
130         android:textSize="18sp" />
131
132     <Button
133         android:layout_width="70dp"

```

```

134         android:layout_height="70dp"
135         android:text="8"
136         android:backgroundTint="#7B1FA2"
137         android:textColor="@color/white"
138         android:layout_margin="4dp"
139         android:textSize="18sp" />
140
141     <Button
142         android:layout_width="70dp"
143         android:layout_height="70dp"
144         android:text="9"
145         android:backgroundTint="#7B1FA2"
146         android:textColor="@color/white"
147         android:layout_margin="4dp"
148         android:textSize="18sp" />
149
150     <Button
151         android:layout_width="70dp"
152         android:layout_height="70dp"
153         android:text="*"
154         android:backgroundTint="#7B1FA2"
155         android:textColor="@color/white"
156         android:layout_margin="4dp"
157         android:textSize="18sp" />
158 </LinearLayout>
159
160 <LinearLayout
161     android:layout_width="match_parent"
162     android:layout_height="wrap_content"
163     android:orientation="horizontal"
164     android:gravity="center">
165
166     <Button
167         android:layout_width="70dp"
168         android:layout_height="70dp"
169         android:text="C"
170         android:backgroundTint="#7B1FA2"
171         android:textColor="@color/white"
172         android:layout_margin="4dp"
173         android:textSize="18sp" />
174

```

```

175     <Button
176         android:layout_width="70dp"
177         android:layout_height="70dp"
178         android:text="0"
179         android:backgroundTint="#7B1FA2"
180         android:textColor="@color/white"
181         android:layout_margin="4dp"
182         android:textSize="18sp" />
183
184     <Button
185         android:layout_width="70dp"
186         android:layout_height="70dp"
187         android:text="."
188         android:backgroundTint="#7B1FA2"
189         android:textColor="@color/white"
190         android:layout_margin="4dp"
191         android:textSize="18sp" />
192
193     <Button
194         android:layout_width="70dp"
195         android:layout_height="70dp"
196         android:text="="
197         android:backgroundTint="#7B1FA2"
198         android:textColor="@color/white"
199         android:layout_margin="4dp"
200         android:textSize="18sp" />
201 </LinearLayout>
202
203 </LinearLayout>

```

6.2. Screenshot

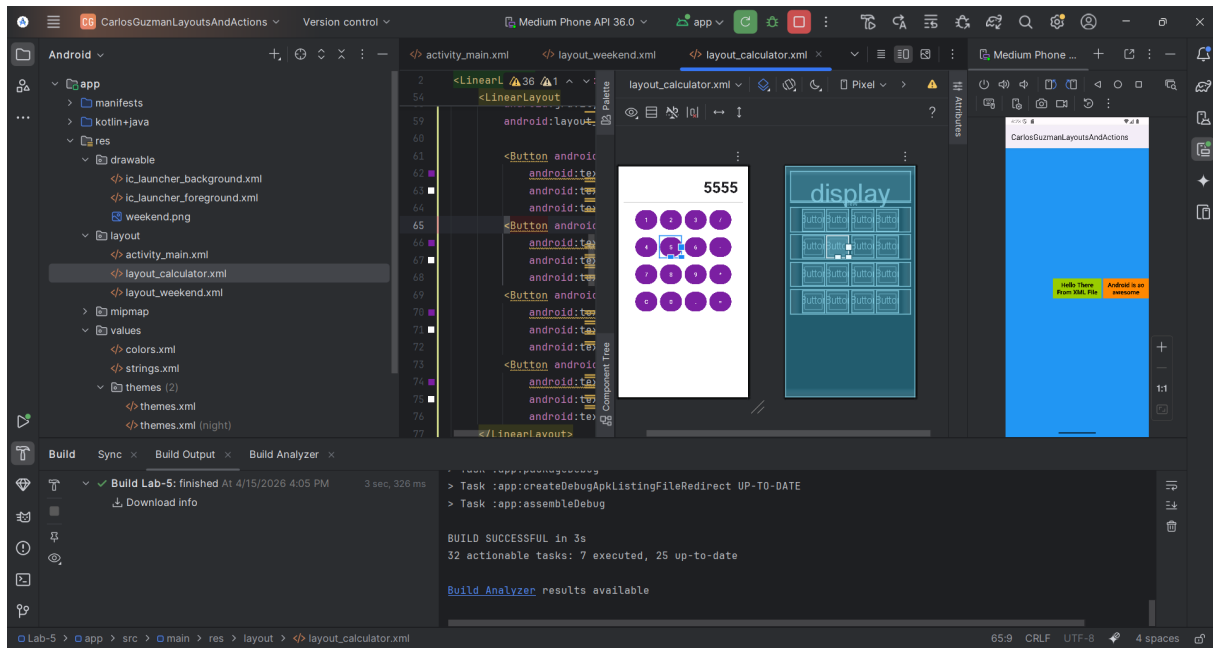


Figure 3: Calculator layout shown in the emulator.

7. MainActivity Configuration

To test each screen in the emulator, the `setContentView()` method in `MainActivity.kt` can be changed to load a specific XML layout.

7.1. MainActivity.kt

```
1 package com.example.carlosguzmanlayoutsandactions
2
3 import android.os.Bundle
4 import androidx.activity.enableEdgeToEdge
5 import androidx.appcompat.app.AppCompatActivity
6 import androidx.core.view.ViewCompat
7 import androidx.core.view.WindowInsetsCompat
8
9 class MainActivity : AppCompatActivity() {
10     override fun onCreate(savedInstanceState: Bundle?) {
11         super.onCreate(savedInstanceState)
12         enableEdgeToEdge()
13
14         // This line was changed depending on the layout being tested.
```

```

15      // setContentView(R.layout.activity_main)
16      // setContentView(R.layout.layout_weekend)
17      setContentView(R.layout.layout_calculator)
18
19      ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main)) { v,
20          insets ->
21          val systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars
22              ())
23          v.setPadding(
24              systemBars.left,
25              systemBars.top,
26              systemBars.right,
27              systemBars.bottom
28          )
29          insets
30      }
31  }
32  }
33  }

```

The layout displayed in the emulator can be changed by replacing the active `setContentView()` line with one of the following options:

```

1 setContentView(R.layout.activity_main)
2 setContentView(R.layout.layout_weekend)
3 setContentView(R.layout.layout_calculator)

```

8. Theme Configuration

The following configuration was used to display the title bar of the application.

8.1. themes.xml

```

1 <resources xmlns:tools="http://schemas.android.com/tools">
2
3     <style name="Base.Theme.CarlosGuzmanLayoutsAndActions"
4         parent="Theme.Material3.DayNight">
5
6         <item name="windowActionBar">true</item>
7         <item name="windowNoTitle">false</item>
8
9     </style>
10 </resources>

```

```
9      </style>
10
11      <style name="Theme.CarlosGuzmanLayoutsAndActions"
12          parent="Base.Theme.CarlosGuzmanLayoutsAndActions" />
13
14  </resources>
```

9. Conclusion

In this laboratory practice, static Android layouts were created using XML files. Different types of containers were used, including `RelativeLayout` and `LinearLayout`. Resources such as colors, strings, and images were also integrated into the project.

Although the project is still being completed, the current progress already demonstrates the main concepts of XML-based interface design in Android Studio. The practice has helped reinforce the use of layout attributes, visual organization, and the relationship between resources and interface files.